

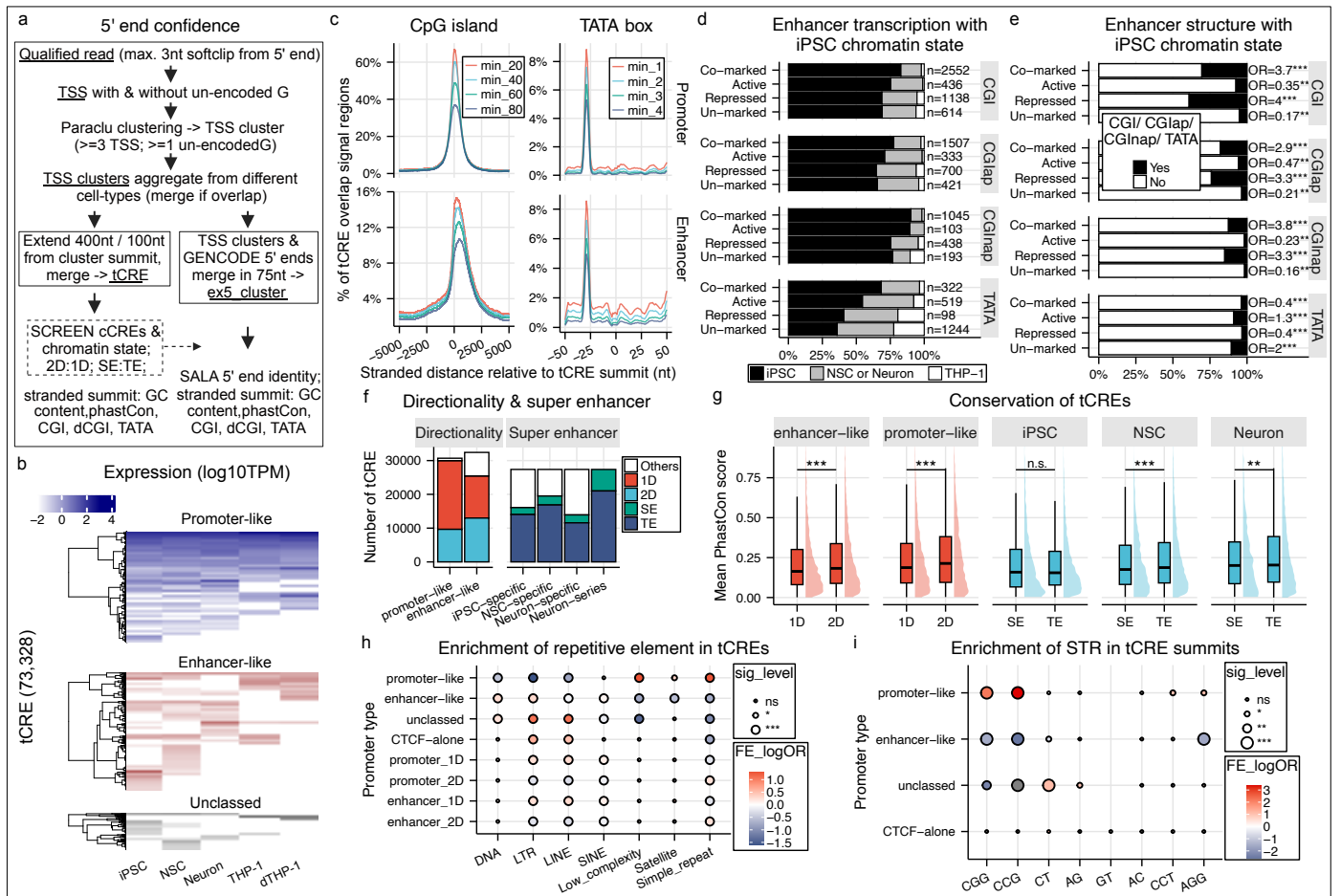


Figure 2 | Identification of genuine TSSs and tCREs from CFC-seq.

a, Flowchart of the SCAFE pipeline for identifying genuine TSS clusters, tCREs, and ex5_clusters from CFC-seq reads. **b**, A total of 73,328 tCREs were identified across five cell types and classified as promoter-like, enhancer-like, or unclassified based on annotation from SCREEN cCRE. Normalized tag count of 10% of these tCRE were visualized in the heatmap. **c**, Distribution of CpG island (CGI) density and TATA box motif score across major-strand tCREs, stratified by promoter-typing. Positions are relative to tCRE summits. **d**, Genomic coordinates of enhancer-like tCREs (major strand only) were intersected with iPSC H3K27Ac and H3K27Me3 CUT&Tag data and grouped into Co-marked (Both signals), Active (27Ac alone), Repressed (27Me3 alone), and Un-marked (Neither). TSS counts ≥ 1 were considered positive in expression. Top panel: percentage of expressed tCREs per enhancer state in CGI-positive (TATA-negative) class; second to bottom panels: CGInap-positive (CGI-positive that are adjacent to promoter), dCGInap-positive (dCGI-positive that are not adjacent to promoter) and TATA-positive (CGI-negative). **e**, Enrichment between enhancer structures (CGI, CGInap, dCGInap & TATA) and chromatin states. Fisher's exact test was used for enrichment (OR: odds ratio). **f**, Number of tCREs classified as unidirectional (1D), bidirectional (2D), super enhancer (SE), and typical enhancer (TE), based on **Extended Data Figure 3a-c**. **g**, Sequence conservation (17-way PhastCons) for 1D vs. 2D and SE vs. TE tCREs, using the averaged conservation score ± 500 nt from tCRE summits. **h**, Enrichment of repetitive elements across promoter types and directionality tested by two-tailed Fisher's exact test. Repetitive element overlap is as in **Extended Data Figure 3m**, where the ones with the longest overlap were selected from "multiple" as representatives. **i**, Enrichment of colocalization of TSS and short tandem repeats (STR) across different promoter types tested by two-tailed Fisher's exact test. Unless specified, statistical tests were two-sided Wilcoxon; $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), n.s. = not significant.